

Lab 3: Component Modelling & Architectural Pattern Selection

Self-Service Coffee Kiosk | Architecture Justification

Architecture Selection

We chose a **Layered Architecture** for the Self-Service Coffee Kiosk, deployed as a modular local application with an isolated external card-payment integration.

Two scenario-specific reasons

1. Fits the single kiosk and its devices. Touchscreen presentation, order rules, menu/pricing data, and printer integration have clear responsibilities. The printer adapter or menu component can be replaced behind its interface without changing the customer workflow.

2. Keeps routine ordering responsive and simple to operate. The three drinks, two sizes, and pricing data can be handled locally; separate network-deployed services for each kiosk function add little value here. Only the required card authorization crosses the kiosk boundary.

Security advantage

The Payment Service is an explicit boundary for credit-card authorization. Send card data only through an authenticated, encrypted processor connection; prefer tokenized or processor-hosted capture; and retain only an authorization result and order reference, never the card number or security code. This reduces card-data exposure, while the payment provider and deployment must still meet applicable PCI requirements.

Performance benefit

Touch input, order validation, and menu/price lookups run locally, avoiding network round trips during ordinary interactions. The external authorization call occurs only when the customer pays, and the printer adapter communicates directly with receipt hardware.

Components and key interfaces

Touchscreen UI calls Order Manager (IOrderAPI); Order Manager reads Menu & Pricing Catalog (IMenuCatalog), authorizes via Payment Service (ICardAuthorization), and submits paid order details to Receipt Printer Adapter (IReceiptJob); the adapter sends the print job to Receipt Printer (IPrinter).